

ABOUT THE PROJECT

► PROBLEM

Caregivers need a safe and simple way to monitor a child's room environment and respond quickly when conditions become unsafe or uncomfortable. This system helps monitor the room and provides added comfort through controlled lighting and fan operation.

♥ SOLUTION

The Smart Child Monitoring and Comfort Control System measures temperature, humidity, light intensity, and sound level in real time. When abnormal conditions are detected, the system can generate alerts and allow remote control of a 12V fan and 12V LED light.

🎯 DESIGN GOALS

- Safe low-voltage operation
- Real-time environmental monitoring
- Fast caregiver awareness through alerts
- Practical comfort control using fan and lighting
- Reliable hardware integration using sensors, protected power, and switched outputs

FINAL PROTOTYPE & CONTACT

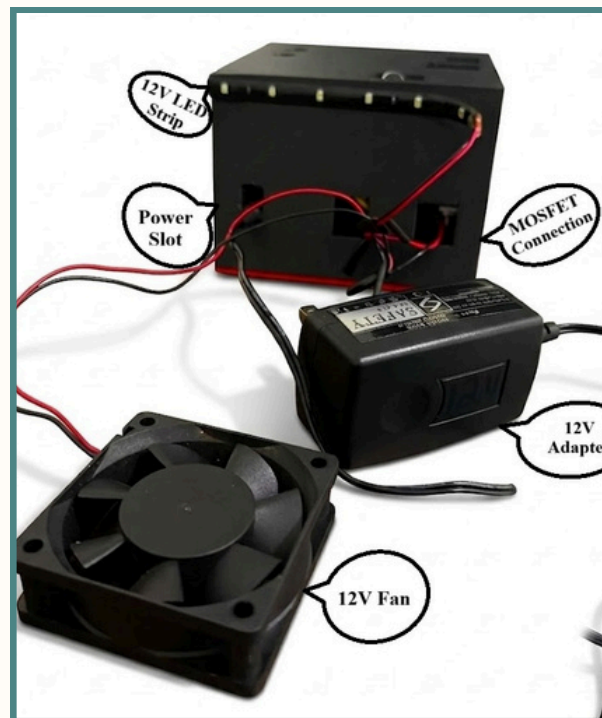


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SAFE LOW-VOLTAGE DESIGN

Monitors room conditions in real time

DESIGN PROJECT
BY NAMAN ARORA

CHILD MONITORING SYSTEM



TEMPERATURE • HUMIDITY •
LIGHT • SOUND • ALERTS

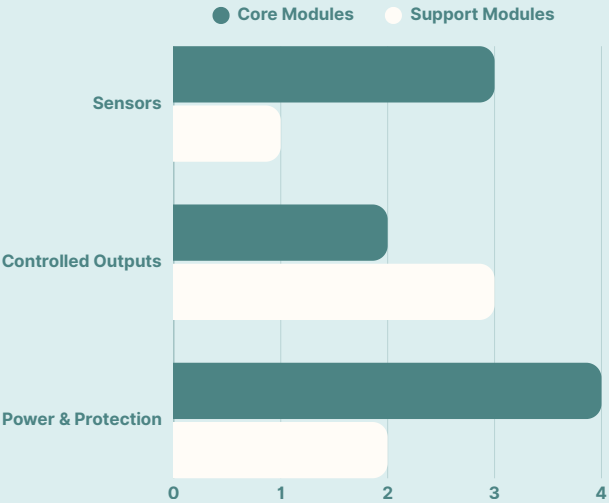
FEATURES & SPECIFICATIONS

KEY FEATURES

Monitors temperature, humidity, light intensity, and sound level in real time. When room conditions become unsafe or uncomfortable, the system can generate alerts, and support remote control of a 12V fan and 12V LED light.

CORE HARDWARE

Built around a Freenove ESP32-S3 CAM, the prototype uses a DHT11 temperature and humidity sensor, a BH1750 light sensor, and an analog sound sensor. The system is powered from a protected 12V input with an inline fuse and LM2596 buck converter, and it drives a 12V fan and 12V LED strip through MOSFET-based switching.



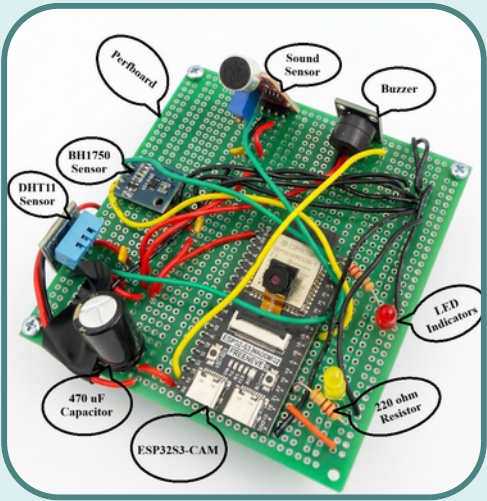
ELECTRICAL SPECIFICATIONS

The system is powered from a protected 12V DC input and uses regulated low-voltage power for control electronics. An ESP32-S3 CAM manages sensing, alert logic and output control for the prototype.

Parameter	Value
INPUT SUPPLY	12V DC Adapter
REGULATED SUPPLY	5V via LM2596 Buck Converter
MAIN CONTROLLER	Freenove ESP32-S3 CAM

PROTOTYPE DETAILS

The prototype integrates temperature/humidity, light, and sound sensing with alert and comfort-control outputs. The system supports a 12V fan, 12V LED strip, buzzer indication, and protected wiring connections for safe low-voltage operation.



OPERATING INSTRUCTIONS



4 MAIN MONITORING INPUTS
TEMPERATURE • HUMIDITY • LIGHT • SOUND



2 COMFORT CONTROL OUTPUTS
12V FAN • 12V LED LIGHT

HOW THE SYSTEM WORKS?

1. Connect the system to the protected 12V DC power input.
2. The ESP32-S3 CAM powers the control logic and reads the room sensors continuously.
3. Temperature, humidity, light intensity, and sound level are compared with preset conditions.
4. If an unsafe or uncomfortable condition is detected, the system can generate an alert and support remote control of the fan and LED light.